MIL-C-60984A (MU)
20 March 1969
SUPERSEDING
MIL-C-60984 (MU)
11 January 1968

MILITARY SPECIFICATION

CARTRIDGE, 40MM, HE, M406; BALL AND SKIRT ASSEMBLY AND OGIVE FOR METAL PARTS FOR

1. SCOPE

1.1 This specification covers the metal parts for the Ball and Skirt Assembly and Ogive for Cartridges, M406 and M381.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-I-45607	-	Inspection Equipment, Supply and Maintenance of

STANDARDS

FEDERAL

FED-STD-151	-	Metals; Test Methods

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-	Quality Assurance Terms and Definitions

FSC: 1310

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MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

8833699 - Ball and Skirt Assembly
8840508 - Ogive

PUBLICATIONS

ARMY

ET-8833699 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts and assemblies.-All parts and assemblies shall comply with all requirements specified on Drawings (dwg.) 8833699, 8840508, and with all requirements specified in the applicable specifications.

3.3 Internal air pressure

3.3.1 Ball and skirt assembly, prior to internal coating-The ball and skirt assembly shall withstand an internal air pressure of fifteen (15) psi minimum (min.) for four (4) seconds, min., without evidence of leakage when tested as specified in 4.4.1.

3.3.2 Alternate procedure

3.3.2.1 Ball assembly, prior to internal coating.-The ball assembly shall withstand an internal air pressure fifteen (15) psi min. for four (4) seconds min., without evidence of leakage when tested as specified in 4.4.1.2.1.

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3.3.2.2 Ball and skirt assembly, after internal coating-The ball and skirt assembly shall withstand an internal air pressure of fifteen (15) psi min. for four (4) seconds, min., without evidence of leakage when tested as specified in 4.4.1.2.2.

3.4 Skirt hardness and compression

3.4.1 Hardness.-The hardness shall be 68 + 10 on the Rockwell E Scale when tested as specified in 4.4.6. This requirement is not applicable when skirt is made from tubing or bar stock.

3.4.2 Compression.-The skirt, when compressed to 3/4 inch in a plane normal to its longitudinal axis, shall show no evidence of cracks when tested as specified in 4.4.5. This requirement is not applicable when skirt is made from tubing or bar stock.

3.5 Workmanship.-All parts and assemblies shall be fabricated and finished in a thorough workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent.

3.6 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (see 6.4)

3.7 First article testing-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

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4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.-The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. (See 6.1) The first article sample shall consist of the following:

a. Seventy (70) ball and skirt assemblies (50 not internally coated).
b. Fifty (50) ball assemblies.
c. Fifty (50) skirts.
d. Twenty (20) of each part.
e. Twenty (20) plugs, prior to forming.
f. Two (2) samples of wire each equivalent in length to the circumference of the notching roll with a minimum of 2 feet in length.

If the alternate procedure for internal air pressure is applied, the initial production sample shall consist of the following:

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a. Seventy (70) ball and skirt assemblies (fifty 50 for internal air pressure).
b. Seventy (70) ball assemblies (fifty (50) not internal coated).
c. Fifty (50) skirts.
d. Twenty (20) of each part.
e. Twenty (20) plugs prior to forming.
f. Two (2) samples of wire each equivalent in length to the circumference of the notching roll with a minimum of two (2) feet in length.

All parts and completed assemblies submitted shall have been produced by the contractor (or furnished by a supplier) using the same production processes, procedures and equipment as will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. A first article quantity, or portion thereof, as directed by the Contracting Offier, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed.-Assemblies and components will be subjected by the Government to any or all the examinations or tests specified in this specification and any or all requirements of the applicable drawings. In addition, any or all of the following examinations and tests will be performed.

4.2.2.1 Hardness of skirt.-The fifty (50) skirts shall be subjected to this test.

4.2.2.2 Inspection.-This paragraph provides guidelines for minimum first article inspection in cases where there is more than one source of supply for one or more components:

4.2.2.2.1 Balls, prior to forming (dwg. 8833705).-Inspect each ball prior to forming for all drawing dimensions and requirements.

4.2.2.2.2 Plugs, prior to forming (dwg. 8833704) .-Inspect each plug prior to forming for all drawing dimensions and requirements.

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4.2.2.2.3 Inserts (dwg. 8833703).-Inspect each insert for all drawing dimensions and requirements.

4.2.2.2.4 Skirt (dwg. 8840512).-Inspect each skirt for all dimensions in main view, except inside spherical radius need not be gaged. Inspect ten (10) parts visually for other features. Test each skirt for hardness (see 3.4.1) and compression (see 3.4.2).

4.2.2.2.5 Ball Assembly (dwg. 8833700).-Inspect each ball assembly for all drawing dimensions and requirements.

4.2.2.2.6 Ball and skirt assembly (dwg. 8833699).-Inspect each ball and skirt assembly in accordance with the provisions of 4.2.2.

The number of initial production samples (70 assemblies each sample - see 4.2) to be submitted shall be equal to the greatest number of suppliers of any one component, and each supplier's components shall be included in at least one initial production sample of ball and skirt assemblies.

NOTE.-In regular production, components which have passed initial production inspection may be mixed in any combination regardless of source of supply to make ball and skirt assemblies.

4.2.3 Rejection.-If any assembly or component fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same

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specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Plug, prior to forming (see dwg. 8833704 covering a detail of dwg. 8833700)

Categories	Defects	Method of Inspection (see 4.3.3.2)	Defect Code (see 6.2)
Critical:	None defined		
Major:	AQL 0.65 Percent		
101.	Thickness from base of notch to opposite surface	Gage	01001
102.	Thickness of material	Gage	01002
103.	Improperly notched	Visual	01003
Minor:			
201.	Evidence of poor workmanship (see 3.5)	Visual	01004

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4.3.2.2 Ball, prior to forming (see dwg. 8833705 covering a detail of dwg. 8833700)

Categories	Defects	Method of Inspection	Defect Code
Critical:	None defined		
Major:	AQL 0.65 Percent		
101.	Thickness of wire	Gage	02001
102.	Depth of notch	Gage	02002
103.	Width of wire	Gage	02003
104.	Incorrect number of notches per inch	Visual	02004
Minor:			
201.	Evidence of poor workmanship (see 3.5)	Visual	02005

4.3.2.3 Ball assembly (see dwg. 8833700 covering a detail of dwg. 8833699)

Categories	Defects	Method of Inspection	Defect Code
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Plug more than .015 from flush	Visual/Scale	03001
Minor:	AQL 1.50 Percent		
201.	Spherical diameter	Gage	03002
202.	Evidence of poor workmanship (see 3.5)	Visual	03003
203.	Countour of ball (plug area)	Visual	03004
204.	Identification mark missing from top of ball (when applicable)	Visual	03005

4.3.2.4 Ball and skirt assembly, prior to internal coating and if the alternate prcocedure is applied, ball and skirt assembly after internal coating (see dwg. 8833699)

Categories	Defects	Method of Inspection	Defect Code
Critical:	None defined		

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Major:	AQL 1.50 Percent		
101.	Hardness of skirt	Test (see 4.4.6)	04001
102.	Distance from curved surface to rotating bands, max.	Gage	04002
103.	Concentricity of pitch diameter of threads with bourrelet of skirt and perpendicularity of fuze seating surface	Gage	04003
104.	Pitch diameter of thread, minimum (min.)	Gage	04004
105.	Diameter of bourrelet	Gage	04005
106.	Diameter of ball end	Gage	04006
107.	Concentricity of outside diameter (for ogive) at fuze end of skirt with bourrelet	Gage	04007
108.	Profile of rotating bands	Gage	04008
109.	Location of rotating bands	Gage	04009
110.	Profile of ogive crimp groove	Gage	04010
111.	Location of ogive crimp groove	Gage	04011
112.	Diameter of rotating band	Gage	04012
113.	Diameter of ogive crimp end	Gage	04013
114.	Height of ball assembly, max.	Gage	04015
Minor:	AQL 1.00 Percent		
201.	Length, max.	Gage	04017
202.	Evidence of poor workmanship (see 3.5)	Visual	04018

4.3.2.5 Ball and skirt assembly (see dwg. 8833699)

Categories	Defects	Method of Inspection	Defect Code
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Weight	Balance	05001
Minor:	AQL 1.50 Percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	05002
202.	Internal protective coating missing	Visual	05003
203.	Incorrect color of skirt	Visual	05004
204.	Protective coating on threads missing	Visual	05005

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4.3.2.6 Ogive (see dwg. 8840508)

Categories	Defects	Method of Inspection	Defect Code
Critical:	None defined		
Major:	AQL 0.65 Percent		
101.	Total length	Gage	06001
102.	Inside diameter	Gage	06002
103.	Large outside radius	Gage	06003
104.	Wall thickness min for required distance	Gage	06004
Minor:	AQL 2.50 Percent		
201.	Diameter of flat	Gage	06005
202.	Protective finish with bare spot	Visual	06006
203.	Color incorrect	Visual	06007
204.	Crack or dent	Visual	06008
205.	Evidence of poor workmanship (see 3.5)	Visual	06009

4.3.2.7 Box, fiber, prior to sealing (see dwg. 9232376)

Categories	Defects	Method of Inspection	Defect Code
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Assembly inverted	Visual	16001
102.	Incorrect number of assemblies	Visual	16002
Minor:	AQL 0.65 Percent		
201.	Fillers missing	Visual	16003

4.3.2.8 Sealed fiber box (see dwg. 9232376)

Categories	Defects	Method of Inspection	Defect Code
Critical:	None defined		
Major:	None defined		
Minor:	AQL 0.65 Percent		
201.	Marking misleading or unidentifiable	Visual	17001

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4.3.3 Testing

4.3.3.1 Dimensional control of notched wire, Code No. 08001) .-In place of the normal sampling associated with the Classification of Defects, two (2) samples of wire, each equivalent in length to the circumference of the notching roll, shall be selected from each coil of wire. One sample shall be taken from the beginning of the coil and the other sample from the end. The first sample shall be dimensionally inspected to qualify the tool for use in production. The second sample shall be dimensionally inspected as a control of the tool during production.

If the qualification sample of the notched wire does not conform to the dimensions as prescribed on the applicable drawing, the tool used for the notching shall not be used in production.

If the control sample of the notched wire does not conform to the dimensions as prescribed on the applicable drawing, tne tool used for the wire shall be removed from production. Further, that portion of wire produced since the last tool control check shall be returned to the contractor for corrective action.

In the event of adjustments made on the notching tool, or the introduction of a new tool, or the return of a reworked tool, the tool must pass the qualification test described above. The contractor may request a change of inspection frequence providing that he presents objective evidence to substantiate the request to the Contracting Officer. When inspection is performed as specified in this Paragraph, the AQL's listed in the Classification of Defects will not apply.

4.3.3.2 Dimensional control of stamped parts, Code No. 09001.-In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least ten (10) parts may be dimensionally inspected to qualify the tool used in the stampling process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control of the tool during production.

If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of

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production since the last tool control check shall be returned to the contractor for corrective action.

All tools removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

4.3.3.3 Dimensional control of molded parts, Code No. 10001 .-In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, and after a curing time for the parts has been approved (See Note 1), a sample of three (3) parts (as molded) from each cavity may be dimensionally inspected to qualify a new or reworked cavity for use in production or as a control of each cavity during production.

The inspection of the parts shall be performed only when the curing time for the parts has been reached. The inspection for control of each cavity during production shall be performed on the above quantities from each cavity after production of each 5000 parts or at the end of each shift, providing the number of parts produced during that shift is less than 5000 parts. Of the three (3) samples, one (1) sample shall be the last part produced. Parts that fail to comply with dimensional requirements after dimensional stability has been attained shall be classed defective.

If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for corrective action.

All cavities formerly removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test as above. The contractor may request a change of inspection frequency providing he presents objective evidence to substantiate the request to the Contracting Officer.

(NOTE 1) - In establishing a curing time, dimensionally check ten (10) parts from each cavity at periodic intervals (e.g. every 30 or 60 mins., etc.) until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data used in determining the curing time shall be sent to Picatinny Arsenal, SMUPA-ND3. If any new change in material,

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cycle time, a reworked cavity, or new cavity is used, a new curing time shall be established and approved.

4.3.3.4 Internal air pressure

4.3.3.4.1 Ball and skirt assembly prior to internal coating (see para. 3.3.1), Critical Defect, Code No. 11001-Each ball and skirt assembly shall be subjected to this test. If any ball and skirt assembly fails to comply with the applicable requirement, it shall be classified defective and removed from the lot.

4.3.3.4.2 Alternate procedure

4.3.3.4.2.1 Ball assembly, prior to internal coating (see para. 3.3.2.1), Critical Defect, Code No. 11002-Each ball assembly shall be subjected to this test. If any ball assembly fails to comply with the applicable requirement, it shall be classified defective and removed from the lot.

4.3.3.4.2.2 Ball and skirt assembly, after internal coating (see para. 3.3.2.2), Critical Defect, Code No. 11003-After completion of test specified in 4.3.3.4.2.1, the ball assembly shall be coated and assembled to the skirt and the assembly shall be subjected to this test. If any ball and skirt assembly fails to comply with the applicable requirement, it shall be classified defective and removed from the lot.

4.3.3.5 Hydrostatic test of ball assembly or ball and skirt assembly (as applicable) (see dwg. 8833700 or 8833699), Major Defect, Code No. 12001).-Eighty (80) ball assemblies or ball and skirt assemblies (as applicable) shall be selected from each lot and subjected to this test. NOTE: Prior to performing this test, and prior to internal coating, the 80 assemblies shall be subjected to an internal air pressure test of 15 psi, min. for 4 seconds, min. applied to the ball assembly only. If during this internal air pressure test, any of the 80 assemblies fail to withstand the 15 psi requirement, it shall be classified defective and replaced with another selected assembly until 80 assemblies have successfully passed the internal air pressure test.

If two (2) or more ball assemblies or ball and skirt assemblies (as applicable) fail to withstand the 1500 psi requirement by rupturing (evidenced by any amount of leakage), the lot shall be rejected.

4.3.3.6 Torque of insert and threads (see dwg. 8833700 and 8833699), Major Defect, Code No. 13001) .-Fifty (50) ball and skirt assemblies shall be selected from each shift's production and subjected to this test. If any ball and skirt assembly fails to comply with the requirements specified on the applicable drawings, the shift's production represented by the sample shall be rejected.

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4.3.3.7 Torque of sphere of ball and skirt assembly (see dwg. 8833699), Major Defect, Code No. 14001.-Fifty (50) ball and skirt assemblies shall be selected from each shift's production and subjected to this test. If any ball and skirt assembly fails to comply with the requirement specified on the applicable drawing, the shift's production represented by the sample shall be rejected.

4.3.3.8 Compression of skirt (see para. 3.4.2), Major Defect, Code No. 15001.-Fifty (50) skirts shall be selected from each shift's production and subjected to this test. If any skirt fails to comply with the applicable requirement, the shift's production represented by the sample shall be rejected.

4.3.4 Inspection equipment.-Equipment Tabulation (ET) number 8833699 and Equipment Lists (EL's) referenced thereon identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3. Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

4.3.4.1 Government rights to documentation .-Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhual, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures

4.4.1 Internal air pressure

4.4.1.1 Ball and skirt assembly, prior to internal coating . The ball and skirt assembly shall be placed in the fixture shown on the Equipment List and the internal air pressure specified in paragraph 3.3.1 shall be applied. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.1.2 Alternate procedure

4.4.1.2.1 Ball assembly, prior to coating-The ball assembly shall be placed in the fixture shown on the Equipment List and the internal air pressure specified in paragraph 3.3.2.1 shall be applied. This test is a non-destructive test. Parts so tested may be returned to the lot.

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4.4.1.2.2 Ball and skirt assembly after internal coating .- The ball and skirt assembly shall be placed in the fixture shown on the Equipment List and the internal air pressure specified in paragraph 3.3.2.2 shall be applied. This test in a non-destructive test. Parts so tested may be returned to the lot.

4.4.2 Hydrostatic test of ball assembly or ball and skirt assembly (as applicable).-The ball assembly or ball and skirt assembly (as applicable) shall be placed in a fixture shown on the Inspection Equipment List and the Hydrostatic pressure specified on dwgs. 8833700 or 8833699 (as applicable) shall be applied. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.3 Torque of insert and threads-The ball and skirt assembly shall be placed in a fixture shown on the Inspection Equipment List and the torque specified on dwgs. 8833699 and 8833700 shall be applied. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.4 Torque of sphere of ball and skirt assembly.-The ball and skirt assembly shall be placed in a fixture as shown on the Inspection Equipment List and the torque specified on dwg. 8833699 shall be applied. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.5 Compression of skirt.-The skirt shall be placed in a fixture and a load supplied in a plane normal to its longitudinal axis to compress the skirt to a dimension specified in para. 3.4.2. Any equipment considered satisfactory to the Contracting Officer may be used in performing the test. Parts so tested shall not be returned to the lot.

4.4.6 Hardness of skirt.-The skirt shall be tested as specified in Federal Test Standard No. 151, Method 243.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level C.-See 5.2.1

5.2 Packing

5.2.1 Level C.-The assemblies shall be preserved and packaged in accordance with dwg. 9232376.

5.3 Marking.-Marking shall be in accordance with dwg. 9232376.

5.4 Shipping.-When shipments from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

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6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Provisions for submission of First Article Samples.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on equipment lists with the inspections listed in this specification. These code numbers shall be listed on drawings of equipment designs submitted by the contractor (see 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Equipment lists.-Equipment lists indicate the availability of inspection equipment designs by showing in the "Part Number" column of the list of inspection equipment (Form SMUPA 1010) the number of drawings for Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.

MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references

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mandatory Army designs, the contractor shall comply with and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor-prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes, and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND8. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished, normally, within one month after receipt.

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6.4 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND3.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1310-A-102

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

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